

HAT-P-19b

R-1358

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-19_230927 · created 2026-07-31T23:25:15+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460214.7567 +/- 0.0045 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.161 +/- 0.049
Transit depth (Rp/Rs)^2	2.61 +/- 1.58 [%]
Orbital Inclination (inc)	89.59 +/- 1.54
Airmass coefficient 1 (a1)	1.037 +/- 0.022
Airmass coefficient 2 (a2)	-0.026 +/- 0.041
Scatter in the residuals of the lightcurve fit is	2.09 %
Best Comparison Star	#2 - [364, 255]
Optimal Aperture	3.94
Optimal Annulus	11.72
Transit Duration (day)	0.104 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.161 ± 0.049 vs 0.1418 (deviation 0.39σ)
Duration fit vs published	0.104 d vs 0.1196 d (deviation 1.4σ)
Mid-time O–C	-4.9 min
Depth SNR	3.2
Residual scatter	2.09 %

Light curve

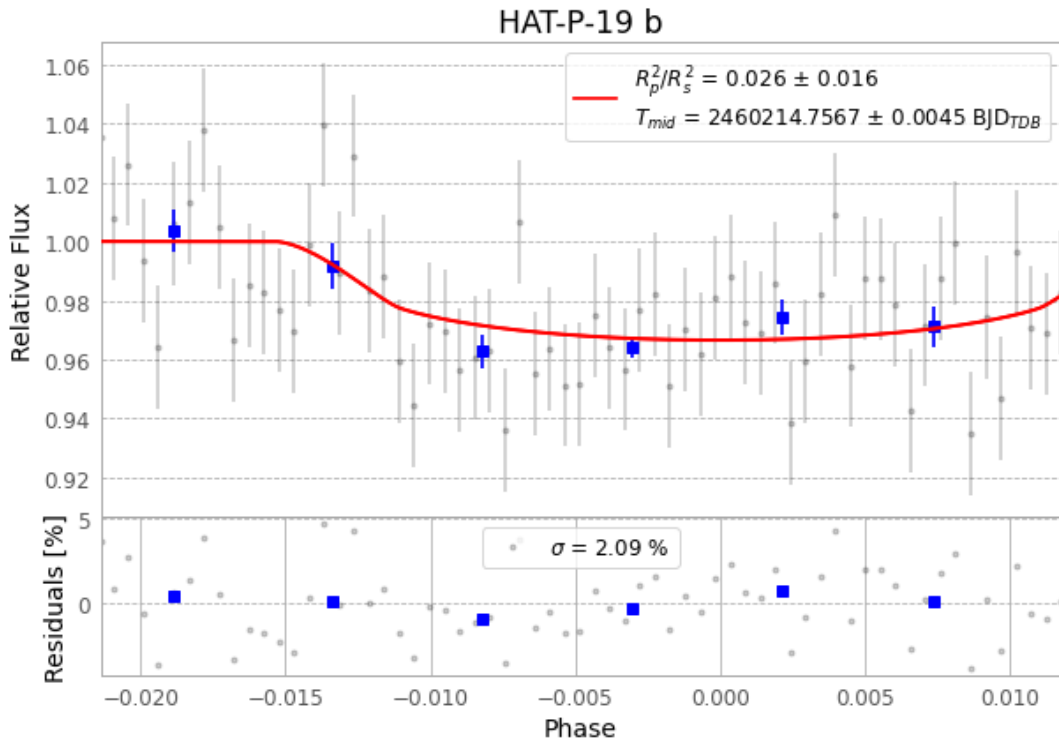


Figure 1 — FinalLightCurve_HAT-P-19 b_2023-09-26.png (SHA-256 71e5f156b6e238e7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [286, 211], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2f2d273bb711bab0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

65 FITS frames (43 MB), HATP-19230927035746.FITS → HATP-19230927070915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-19-230927.log (ae2bad613c6b...), FinalLightCurve_HAT-P-19 b_2023-09-26.png (71e5f156b6e2...), FinalLightCurve_HAT-P-19 b_2023-09-26.csv (29077df86b73...)

Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 23:25Z.